



Magnetic elements

Several lanthanides are very magnetic, and tiny amounts of them are added to iron and nickel magnets to boost their strength. Lanthanide-rich magnets are used to read and write computer memory and to make pickups – devices in electric guitars that convert the vibrations of the strings into electric signals.

The pickups on this guitar use magnets made of the lanthanide samarium.

Electric guitar

Light emission

Lanthanides are crucial in the production of coloured light in light-emitting diodes (LEDs). Blue LEDs get their colours from chemicals containing europium, while green LEDs use chemicals with terbium.

**Blue LEDs glow on
Dubai's Meydan Bridge**

